

Slam - Video Navigation - Exercise 1

Students: Ron Keter, Yehonatan Hirschel

Git Repository: github.com/yehonatan1305/SLAM-VN

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Question 1



Figure 1: Keypoints Detected in Left and Right Stereo Images (ex1.py lines 53-66)

Question 2

Descriptor output generated in ex1.py lines 70-72. output:

```
code/sensor-projects/slam-vn/code/ex1.py
The descriptors of the two first features in image 0 are:
[[ 96  44 229 101  93 111 225  42 123 166 170  9 242  61  85  51  9 244
 123  32  40 255 249  78  93  30  85  52  65 104  86 106]
 [108 165 152 170 157  45  70 201 135  31 173  25 198 227  72 208 217 202
 205  58  26  35 202  51 235 245  5 125 146 123 218  2]]
The descriptors of the first two features in image 1 are:
[[248  89 186  44  73 127  81  94  68 236 230  9  62 181  35 219  72 220
 234  64  12 251 241  40 211 225 112  57 114 125  46  90]
 [ 48  36 240 101  87  47 242  74 127  33 250  10  75 121  20 144 187 221
  70  33 138 178  89 207 183 222  5  6 237  74  82 130]]
```

Figure 2: Feature Descriptors from ORB Algorithm (ex1.py lines 69-73)

Question 3

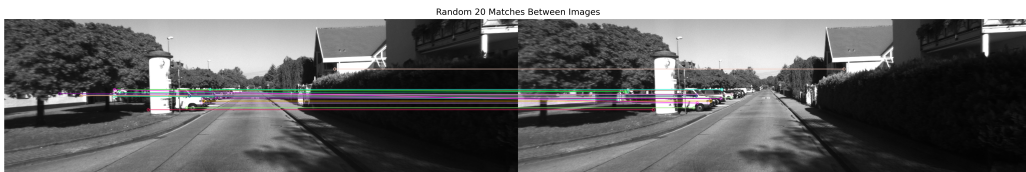


Figure 3: Random 20 Matches Between Stereo Image Pair (ex1.py lines 76-90)

Question 4

Results from running the algorithm:

number of matches before significance test: 500

number of matches discarded: 355

For our implementation of the significance test, we selected a ratio threshold of 0.75, with an upper bound rejection threshold of 0.80. This decision was based on several factors:

- The Wikipedia link provided that Lowe threshold (0.8) rejected 90% of false matches and discarding less than 5% of the correct matches. Therefore, we wanted a threshold that close to this value.
- Visually examining the output pictures.
- Viewing open-CV tutorial and official documentation, usually values of 0.70 and 0.75 used as a threshold. That further strengthens and supports the choice we made.

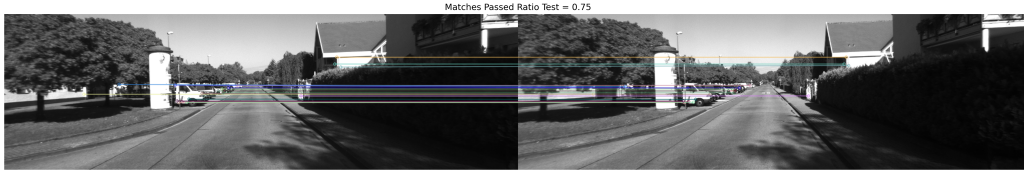


Figure 4: Good Matches Passing Ratio Test (ex1.py lines 99-108)

To present correct matches that failed the significance test, we followed this method:

- We performed a significance test using a higher threshold, making the test less strict.
- We identified keypoints that failed the test with the lower threshold (0.75) but passed with the higher threshold. These keypoints are potentially valid matches that were rejected by the stricter threshold.
- From these keypoints, we selected the 5 matches with the lowest descriptor distances. Visualizing these matches confirmed that they are indeed valid.

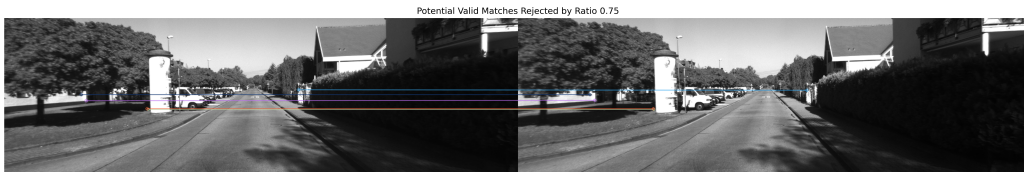


Figure 5: Potential False Negatives (ex1.py lines 110-127)